Andhra University College of Engineering

Computer Science and Engineering | Course Code: CS301

Operating Systems - Operating Systems Process Management & CPU Scheduling N

Overview: Covers process control blocks PCB, CPU scheduling algorithms FCFS, SJF, Round Robin,

I. UNIT-WISE SYLLABUS BREAKDOWN & HIGH-YIELD TOPICS

Unit 1: Process Concept & CPU Scheduling
Process states, PCB, Context switching, FCFS, SJF, Priority, and Round Robin scheduling

Unit 2: Inter-Process Communication & Synchronization
Critical section problem, Peterson solution, Semaphores, Monitors, and Classic IPC problems

Unit 3: Deadlocks & Handling Strategies
Resource allocation graphs, Banker algorithm for deadlock avoidance, detection, and recovery

Unit 4: Memory Management & Virtual Memory
Paging, Segmentation, Page fault handling, FIFO, LRU, Optimal page replacement algorithms

II. HIGH-YIELD SEMESTER EXAMINATION QUESTIONS

1. Derive the average waiting time and turnaround time for SJF and Round Robin scheduling algorithms.
2. Describe the Bankers algorithm for multi-resource deadlock avoidance with safety algorithm proof.
3. Explain demand paging and the working principle of the Inverted Page Table architecture.

III. PRESCRIBED TEXTBOOKS & ACADEMIC REPOSITORIES

* Abraham Silberschatz, Peter B. Galvin - Operating System Concepts Wiley 10th Ed
* Andrew S. Tanenbaum - Modern Operating Systems Pearson

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